

XENO

Engineering Take-Home Assignment

Build an AI-Native Mini CRM for Reaching Shoppers

Hello, and thank you for your interest in joining Xeno as an engineer!

Xeno helps consumer brands reach their shoppers in meaningful, data-driven ways — organising customer data, deciding who to talk to, and running personalised campaigns across channels like **WhatsApp, SMS, Email and RCS**. This assignment gives you a taste of the kind of work we do every day, and gives us a window into how you think, build, and ship.

This is the same assignment for both our SDE and FDE roles. We want to see how you approach an open-ended product problem, end to end.

The Challenge

Build and deploy an AI-native Mini CRM that helps a brand intelligently reach its shoppers.

Imagine a Direct-to-Consumer or retail brand — say a fashion label, a coffee chain, or a beauty brand — with a base of shoppers and their purchase history. Your product should help that brand **decide who to talk to, what to say, and reach them over messaging channels** like WhatsApp, SMS, Email and RCS.

At a minimum, the system you build should be able to:

- **Ingest data** — take in customers and their orders, and store them.
- **Segment shoppers** — let the marketer (or your AI) carve out audiences from that data based on behaviour and attributes.
- **Send personalised communications** — dispatch tailored messages to a chosen audience through a channel service (see below).
- **Surface communication performance insights** — track and present how communications performed (e.g. sent, delivered, failed, opened, read, clicked, order came because of this communication) at the campaign and/or audience level.

Beyond that, **how you build it is deliberately up to you**. We are not handing you a list of modules and screens like a traditional spec. Figuring out exactly what to build — and what NOT to build — is part of what we are evaluating.

What we mean by “AI-native”

The word “intelligent” above is intentional. We want to see AI woven into the product itself, not bolted on. There is no single right shape for this — and we expect different candidates to make very different bets. For example:

- **A classic product interface** where AI assists the marketer at key steps (e.g. drafting messages, suggesting segments).
- **A chat-first experience** where the marketer describes intent in natural language and the product responds.

- **An AI that helps the marketer think, decide and act** — surfacing the right audience, recommending the message, choosing the channel.
- **A true AI agent** that takes a broad goal brainstormed with the marketer and then executes the campaign end to end.

These are illustrations, not requirements. Pick a point of view and commit to it.

The Channel Service (Important)

Do **not** integrate any real messaging provider. Instead, we want you to **stub the channel yourself** as a separate service, and model the full lifecycle of a communication. Concretely:

- Your CRM exposes a **send API**. When a campaign goes out, the CRM calls a **separate stubbed channel service** with the communication details (recipient, message, channel).
- The channel service **does not actually deliver anything** — it simulates outcomes. Asynchronously, it calls back into a **CRM receipt API** with what “happened” to each communication: **delivered, failed, opened, read, clicked**, and so on.
- The CRM ingests these callbacks and updates the state and stats of each communication accordingly.

This two-service, callback-driven loop is deliberate — it’s close to how channel delivery and engagement tracking actually work, and how you model it tells us a lot. How you handle volume, ordering, retries, and failures in this loop is exactly the kind of system-design thinking we want to see.

Scope: What This Is (and Isn’t)

To keep everyone pointed in the right direction:

- **This IS** a CRM for **reaching shoppers / consumers** — a marketing and engagement tool, in the spirit of what Xeno does today.
- **This is NOT** a sales or customer-support CRM for managing deals, pipelines, leads or tickets (think Salesforce, Attio, Clarify). Please don’t build that.

Within that boundary, the depth, the data model, and the “intelligence” are all your calls to make. Use realistic, well-simulated data — you don’t need real customers or real orders to demonstrate this well.

What We’re Evaluating

Read this carefully — it should guide every decision you make.

What we evaluate	What we’re looking for
Build & deploy (table stakes)	A live, hosted, working product with a walkthrough video. This is the baseline expectation, not a differentiator — you stand out on everything below.
Creativity in scoping	How sharply you figured out WHAT to build. The brief is intentionally open. The best submissions make bold, opinionated product choices rather than building everything shallowly.

What we evaluate	What we're looking for
AI-native development	How well you leveraged AI to build. We want to see an AI-native workflow — how you direct, review, and integrate AI output, not whether you used it.
Code quality & structure	Clean, readable, well-organised code with sensible structure. We will read your code and ask about it.
System design & scalability	Your scale assumptions and the tradeoffs you consciously made. We care about your reasoning more than a perfect architecture.
Thought clarity & communication	How clearly you think, present, and explain — in your video and in your decisions.

In short: a working, deployed product is the baseline. You stand out on the creativity of what you chose to build, how AI-native your approach and your workflow were, the quality of your code and system thinking, and how clearly you can present it all.

Deliverables

1. **A hosted, working product.** A public URL we can open and use.
2. **Your code.** A repository (e.g. GitHub) we can read. Your interviewers will review it and ask detailed questions about your choices and logic, so make sure it's your own work.
3. **A short walkthrough video, narrated by you.** Around 5–6 minutes. A structure that tends to work well is below — treat it as a suggestion, not a mandate. What matters is that you cover the product, the thinking, and the build clearly.

Section	Suggested time	What to cover
Product intro	~0.5 min	What you built and why. The problem you chose to solve.
Functional demo	~1.5 min	End-to-end product demo. Show it actually working.
Technical architecture	~1 min	Your architecture diagram and the reasoning behind each major decision.
Code walkthrough	~1 min	Walk us through the structure and a couple of key parts of your code.
AI-native workflow	~1 min	How you leveraged AI while building — how AI-native your dev workflow was.

A Few Ground Rules

- **Use AI freely.** We actively want you to. We are an AI-native company and we expect our engineers to be AI-native too. Just be ready to explain and defend everything you ship — we will ask.
- **Make your tradeoffs explicit.** State your scale assumptions and what you consciously chose not to do. “I’d do X at scale but did Y for this scope” is a great answer.
- **Pick your own stack.** Use whatever lets you build and ship best. We have no stack requirement for this assignment.
- **Originality matters.** Your code will be closely reviewed and discussed live. Understand everything in your submission.

Timeline & Submission

Submissions are due by **12 PM on June 15, 2026**. Submit your hosted URL, repository link, and walkthrough video via the following submission form.

[Submission link for FDE candidates](#)

[Submission link for SDE candidates](#)

We’re genuinely excited to see what you build. Surprise us. ✨